DS2002 Midterm Project | Fall 2026

Walmart Hurricane Analytics

What Florida shoppers bought before Hurricane Helene (September 2024)

Team HEJ: Evie McMahan, Julia McPhillips, Honor Dold

Pop-Tarts surged about 5x, and our cleaning choices changed our answers more than our analysis did

The data

The raw data had a lot of problems

60 store IDs for 12 stores

FL-239, FL239, FL_239 and fl-239 are all the same Orlando store. The store file also listed Orlando and Kissimmee twice.

4 SKUs for one Pop-Tart

Strawberry Pop-Tarts were split across PT-12, POPTART-STRAW, PT_STRAWBERRY and SKU#459812. Overall, 45 SKUs were really 20 products.

886 missing timestamps

Missing values represented 3.5% of the transactions and were written 5 different ways: NULL, N/A, NaN, nan and blank.

Other issues

30 duplicate rows, 130 negative quantities, prices stored as text, and 37 category names for 10 categories.

The pipeline

How we cleaned and joined the data

- Our cleaning function, clean_all(), runs 11 steps, and we logged 32 decisions
- Transactions went from 25,562 rows to 25,532 rows
- We mapped 45 SKUs down to 20 products
- We pulled weather data, including rain, max wind, and lowest pressure, from the Open-Meteo Historical API
- We used Eastern time for our weather data so it matched the sales dates.
    - If we used UTC, 8 of every 48 hourly readings would have matched the wrong day
- The join kept all 30 sales days, and the unit total did not change

Decision 1: timestamps

We kept the 884 rows with no timestamp.

Dropping them would remove real sales, but if we filled in a date we would just be guessing. We used these rows in our totals but left them out of anything by date. (The raw file had 886, and 2 were duplicate rows we removed.)

Decision 2: negative units

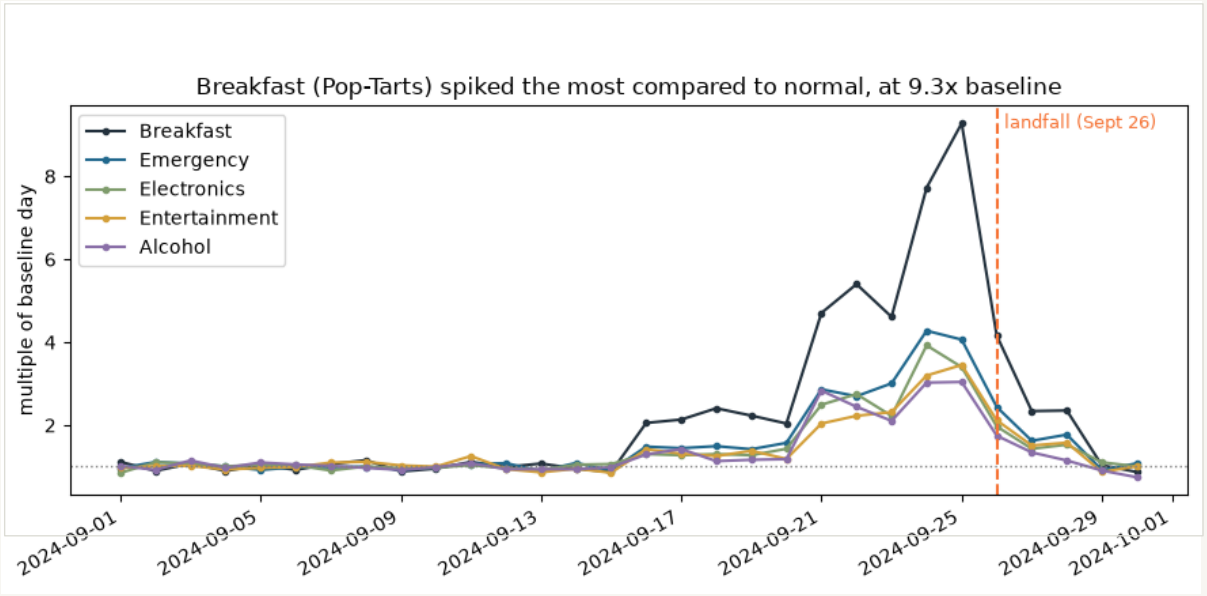
We flipped 130 negative quantities.

None matched a return, so we treated them as sign typos. Total units went from 49,343 to 49,825.

Q1 Demand Surge

Breakfast (Pop-Tarts) had the biggest surge at +414%

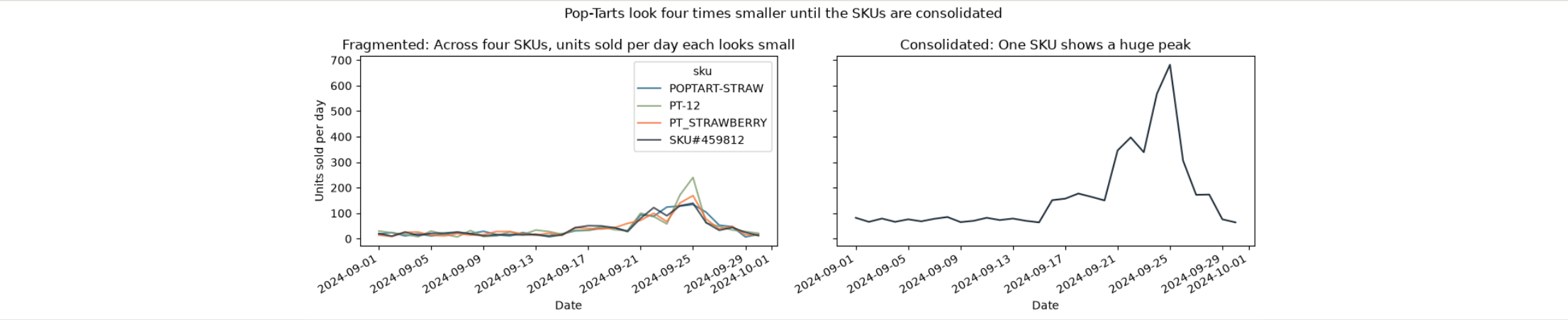
- Breakfast went from 73.5 to 378 units a day (5.1x, or +414%)
- Emergency +184%, Electronics +150%, Entertainment +126%, Alcohol +126%
- Snacks only went up 25%, so people were not buying more of everything
- Sales started rising Sept 16, peaked Sept 24 and 25, and dropped on landfall day



Chosen time periods:
Baseline: Sept 1 to 15. Surge: Sept 19 to 25. Landfall: Sept 26

Q2 The Pop-Tarts Effect

Combining the 4 Pop-Tart SKUs changed the top product



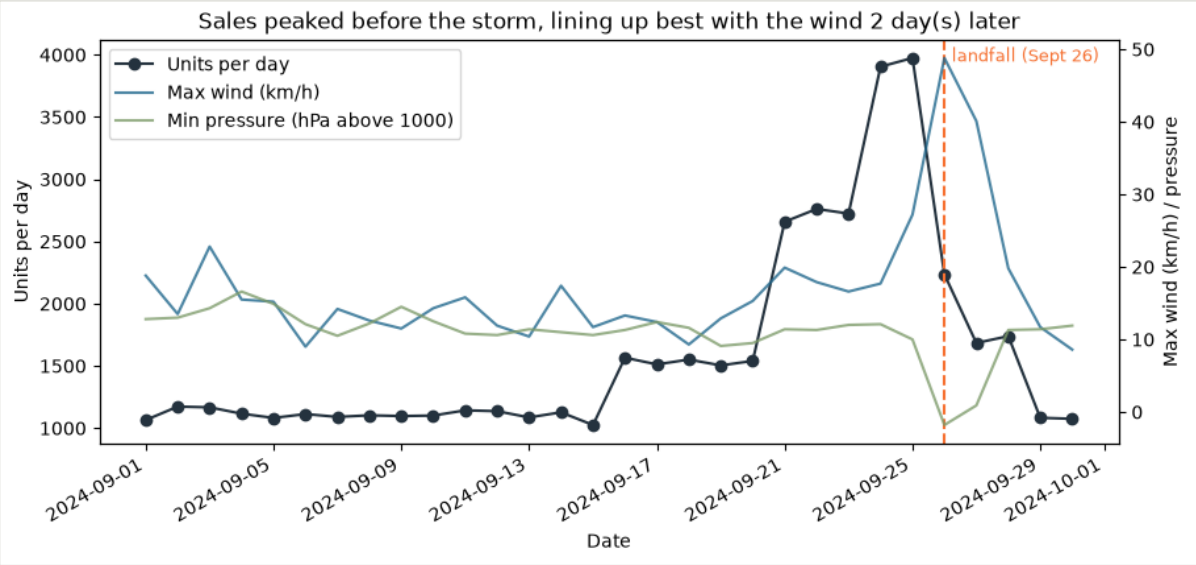
	Before combining	After combining
Units per day in the surge	103 (PT-12)	378
Rank in the surge	4th, behind 3 water SKUs	1st, ahead of water (321)
September total	1,385	5,218 (3.8x more)

- Peak: 682 units on Sept 25, 9.3x the baseline
- Without combining the SKUs, Walmart would not order enough of the top product

Q3 Weather Correlation

People stocked up about 2 days before the worst wind

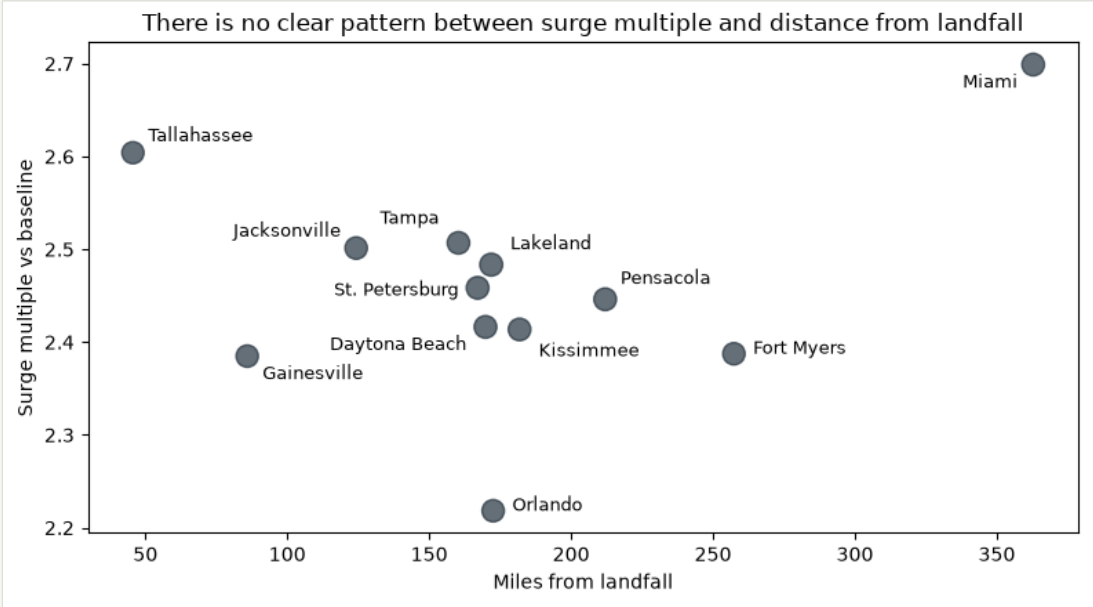
- Same day: wind r = 0.40, pressure r = -0.22, rain r = -0.26
- Weather 2 days later: wind r = 0.82, pressure r = -0.76
- Every category was strongest at 2 days (Health r = 0.85, Snacks r = 0.67)
- Rain stayed weak at every lag
- People stocked up about 2 days before the worst weather
- Note: weather is from Orlando only, and the 2-day correlations use 28 days



Q4 Stores and Geography

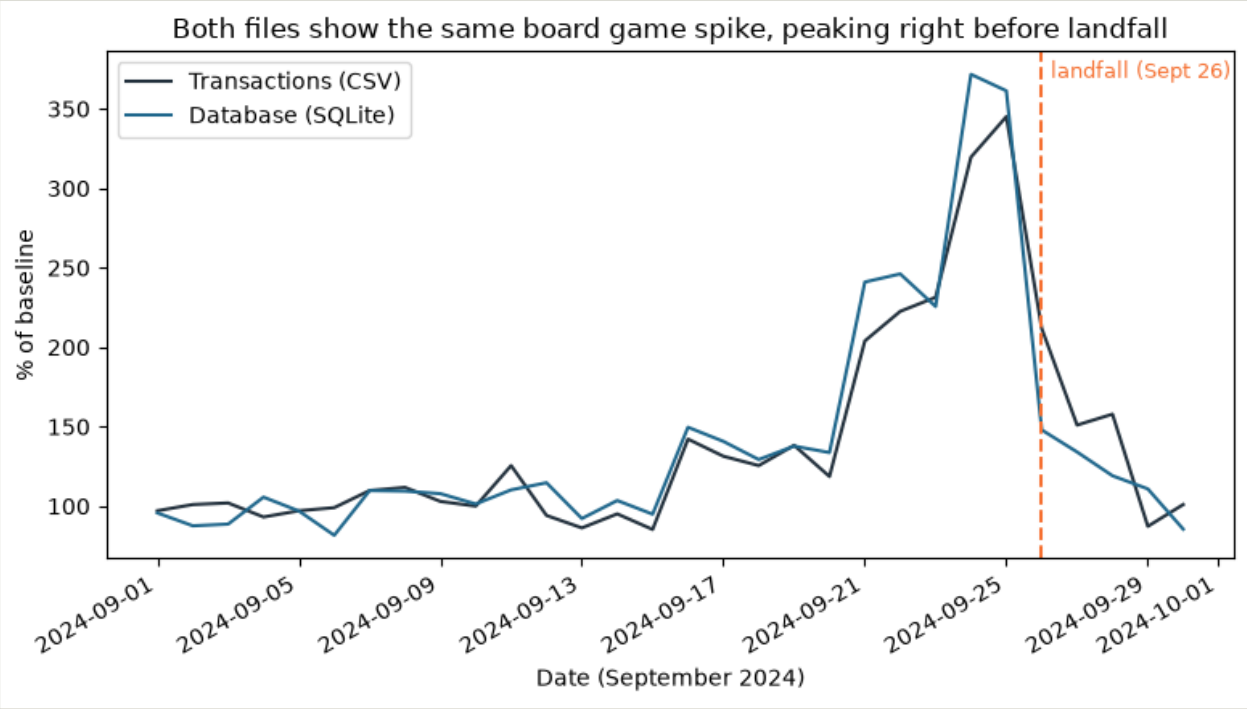
Distance from landfall did not explain the surge

- All 12 stores surged between 2.2x and 2.7x their baseline
- Distance vs. surge: r = 0.23 across 12 stores (farther stores surged slightly more)
- Every store peaked on Sept 24 or 25
- Miami was the highest (2.70x) and Orlando was the lowest (2.22x)
- Miami's Pop-Tarts surged 7.2x vs. 4.0x to 5.8x at other stores, but on a baseline of only about 5 units a day



Q5 Board Games: Signal or Artifact?

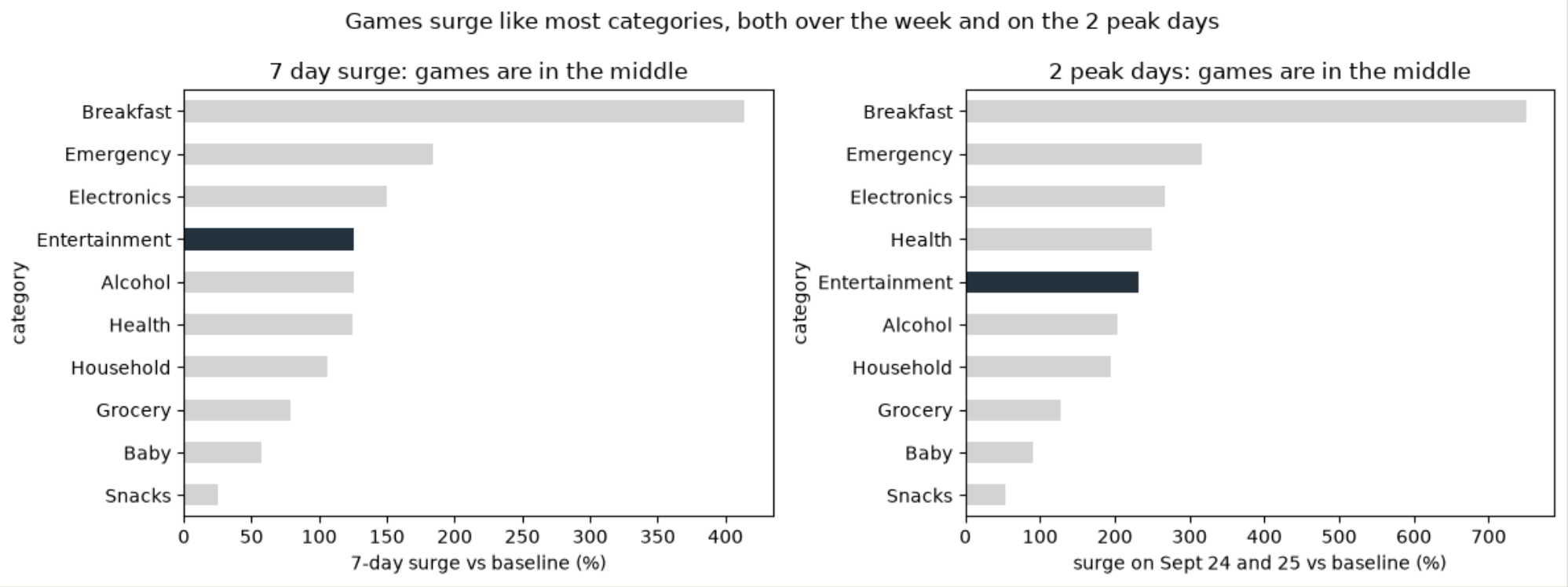
The board game surge is real, but +250% is only the 2 peak days



- On Sept 24 and 25: +232% in the transactions and +267% in the database
- Over the full 7 days: +126% in the transactions and +145% in the database

Q5 Board Games: Signal or Artifact?

Entertainment sat in the middle of the 10 categories



- Games surged like most categories, both over the 7 day surge and on the 2 peak days
  - Entertainment ranked 5th of 10 categories over the peak days, and 4th over the 7 days
- Breakfast, Emergency and Electronics rose more, and Snacks rose the least
- So it is a normal storm peak, not something special to games

Q5 Board Games: Signal or Artifact?

We checked for artifacts and did not find any

What we checked	What we found
Both data sources	The transactions file and the database move together day by day (r = 0.96)
Category lumping	Both products went up (Monopoly +164%, Playing Cards +98%) and all 12 stores surged
SKU miscoding	Monopoly SKUs all sell for about $20 and card SKUs for about $5. No game names outside Entertainment
Inventory reporting lag	Sales and stock both changed on Sept 21, so no lag
A few big orders	Transactions per day went from 57.9 to 133.3, with about the same units per purchase (1.8 vs. 1.7)

Recommendation

What Walmart should do before the next storm

1. Stock every Florida store, not just the ones closest to the storm path (Q4: every store surged 2.2x to 2.7x)
2. Order about 5x normal Pop-Tarts for the week before landfall, and up to 9x for the 2 days before (Q1 and Q2: 5.1x surge, 9.3x peak)
3. Give Emergency items the most truck space (1,011 units a day in the surge)
4. Have stores fully stocked 5 days before landfall, since sales jump then and peak 2 days before the worst wind (Q3)
5. Order about 2.6x normal Monopoly and 2.0x normal playing cards, but after water, batteries, flashlights and Pop-Tarts (Q5: Entertainment ranked 5th of 10 on the peak days)

Limits

What our results cannot tell Walmart yet

- Weather comes from Orlando only, so we could not compare weather store by store
- We studied one storm, so the surge sizes and the 2 day lead may not repeat
- The 2 day correlations use only 28 days of data
- We had no cost or margin data, so we could not size the cost of overstocking
- **Next step:** repeat this on several storms and add cost data

Thank you

Questions?